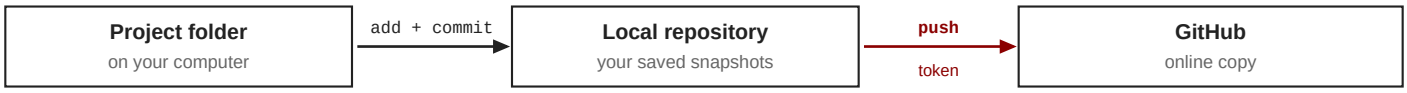
Setting Up Git and GitHub for the First Time

Who this is for

Windows users with no coding experience who need to put a project on GitHub. You will install Git, link it to your GitHub account, and upload your first project. Type each command exactly as shown, then press **Enter**.



Key terms

Terminal where you type commands	**Git Bash** a terminal that comes with Git for Windows; you run Git commands in it	**Git** tracks changes to your files
GitHub a website that stores your projects online	**Repository** a project folder Git tracks	**Commit** a saved snapshot of your files
Push sends your commits to GitHub	**Personal access token** a code that works like a password	**Branch** a line of saved work; yours is called main

Install Git

- Open **PowerShell** (Start menu, type **PowerShell**).
- Type `git --version`. If a version number appears, skip to the next section.
- If you get an error, type:

```
winget install --id Git.Git -e --source winget
```

- Close **PowerShell**. Open **Git Bash** from the Start menu and use it for every step from here on.
- Run `git --version` again. You should see a version number.

Fig. 1 Typing git --version

Installed:

```
$ git --version
git version 2.50.1
```

Not installed:

```
> git --version
The term 'git' is not recognized
```

Make a GitHub Account

- Go to **github.com** and click **Sign up**.
- Confirm your email, then write down your username and email.

Heads up
Use the **same email** for Git and GitHub, or your commits will not show under your account.

Tell Git Who You Are

- Type these three commands, pressing **Enter** after each. Use your GitHub email.

```
git config --global user.name "Your Name"
git config --global user.email "you@example.com"
git config --global init.defaultBranch main
```

- Check them with `git config --global --list`. You should see your name, email, and `main`.

Caution: Copy commands exactly. One typo can break them.

Fig. 3 Your saved settings

```
$ git config --global --list
user.name=Your Name
user.email=you@example.com
init.defaultbranch=main
```

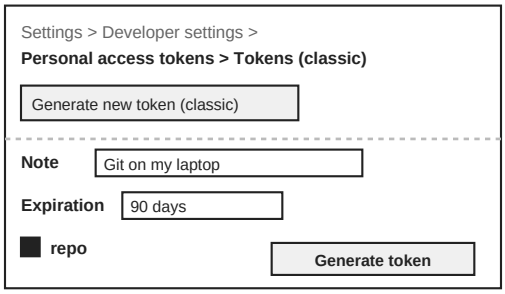
Get a Login That Works with Git

- On GitHub, click your **profile photo**, then **Settings**, **Developer settings**, **Personal access tokens**, **Tokens (classic)**, and **Generate new token (classic)**.
- Name it, set it to expire in **90 days**, check **repo**, and click **Generate token**.
- Copy the token right away and save it somewhere private. GitHub only shows it once.
- In **Git Bash**, type this so Git remembers your login:

```
git config --global credential.helper manager
```

Caution: Never share your token or post it online. It works like a password. It also expires, so you may need a new one later.

Fig. 4 The token page (simplified)



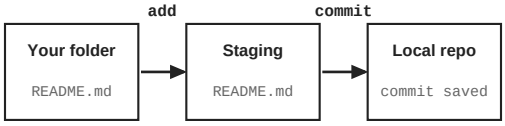
Make Your First Repository

- On GitHub, click **+**, then **New repository**. Name it **my-first-repo**, leave the **README** box unchecked, and click **Create repository**.
- Copy the **HTTPS** link it shows you.
- In **Git Bash**, type these one at a time:

```
mkdir my-first-repo
cd my-first-repo
git init
echo "# My First Repo" > README.md
git add README.md
git commit -m "Add README"
```

- You should see `1 file changed` .

Fig. 5 How a file gets saved



Push to GitHub

- Connect your folder to GitHub. Use the link you copied:

```
git remote add origin YOUR-LINK
git push -u origin main
```

- When asked, type your GitHub username. For the password, paste your **token**, not your account password. Nothing shows while you paste. Press **Enter**.
- Refresh your repository page on GitHub. `README.md` should be listed.

Fig. 6 The push, then your repo online

```
$ git push -u origin main
Username for 'https://github.com':
Password for ...@github.com:
To github.com/you/my-first-repo.git
 * [new branch]  main -> main
```

Troubleshooting

Error	Cause	Fix
`Authentication failed`	Wrong or expired token, or you used your password	Make a new token and paste it as the password
`'git' is not recognized`	Git is not installed, or the window was not reopened	Redo Install Git
`remote origin already exists`	You ran the remote command twice	Type `git remote set-url origin YOUR-LINK`
`failed to push some refs`	GitHub repo was created with a README	Delete the repo on GitHub and remake it with no README
`Please tell me who you are`	Your name and email were never set	Set them again, then commit again

Sources: Git documentation (git-scm.com/doc); GitHub Docs, "Managing your personal access tokens" and "Caching your GitHub credentials in Git" (docs.github.com).